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SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026
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Course Code	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/I	Regulation	R23
Date and Day of Assignment	Week 7 - Thursday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All Batches
Assignment Number: 14.4 (Present assignment number) / 24 (Total number of assignments)			
Q.No.	Question	Expected Time to complete	
1	Lab 14: Web Frontend Development – AI-Assisted HTML/CSS/JS Generation	Week 7	

	Lab Objectives • Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance. • Apply responsive, accessible, and interactive design principles. • Create practical UI components for real-world web applications. • Use AI to optimize layout, UX, and performance. Lab Outcomes • Applying AI assistance to frontend development tasks • Understanding how HTML, CSS, and JavaScript work together • Improving responsiveness, interactivity, and usability • Evaluating and refining AI-generated frontend code	
	Task 1: Responsive Homepage Layout for a Startup Website Scenario You are designing a basic homepage for a startup company that needs to look good on desktops, tablets, and mobile devices. Task Description • Create a web page structure using HTML that includes: ◦ A header with navigation links ◦ A main content section ◦ A footer with basic information • Use AI assistance to generate responsive CSS using media queries. • Ensure the layout adjusts smoothly across different screen sizes. Expected Output • A responsive web page where: ◦ Header, content, and footer are clearly separated ◦ Layout adapts correctly to mobile, tablet, and desktop views ◦ Code is clean and well-structured	

StartUp

HomeServicesAboutContact

Welcome to StartUp

We build modern digital solutions for growing businesses.

Learn More

Our Services

Web Design

Beautiful responsive websites tailored to your brand.

Development

Robust applications built with modern technologies.

Marketing

Strategies that help you reach the right audience.

Task 1: Responsive Homepage Layout for a Startup Website

Scenario

You are designing a basic homepage for a startup company that needs to look good on desktops, tablets, and mobile devices.

Task Description

- Create a web page structure using HTML that includes:
 - A header with navigation links
 - A main content section
 - A footer with basic information
- Use AI assistance to generate responsive CSS using media queries.
- Ensure the layout adjusts smoothly across different screen sizes.

Expected Output

- A responsive web page where:
 - Header, content, and footer are clearly separated
 - Layout adapts correctly to mobile, tablet, and desktop views
 - Code is clean and well-structured python code without comments

Created responsive homepage with embedded HTML and CSS

Created + index.html

Created + styles.css

Created two files in startup-homepage :

index.html — Page structure with:

- Header: sticky nav bar with logo, navigation links, and a hamburger

+ index.html

Describe what to build next

+ Claude Opus 4.6

Our Services

Web Design

Beautiful responsive websites tailored to your brand.

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Robust applications built with modern technologies.

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Strategies that help you reach the right audience.

About Us

We are a passionate team helping startups succeed in the digital world.

Get In Touch

Email us at hello@startup.example

Task 2: Interactive Call-to-Action Button

Scenario

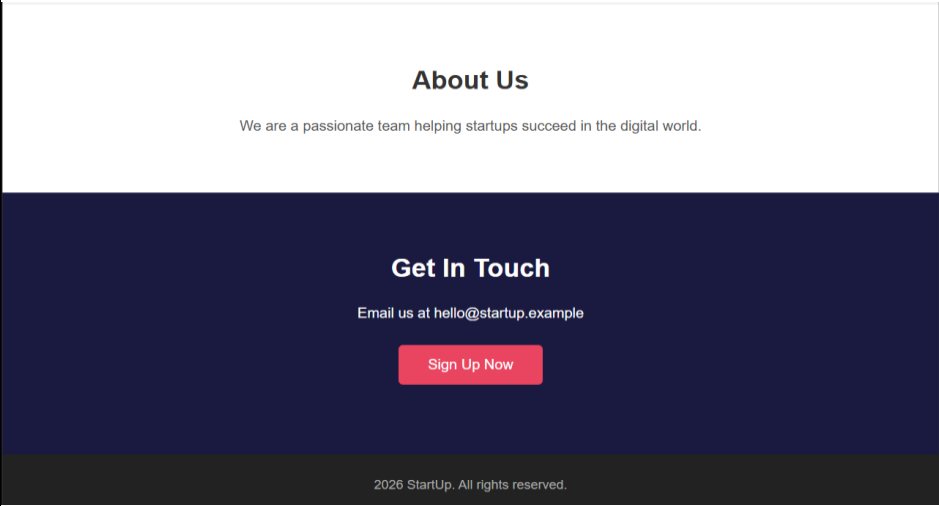
A marketing page requires a **call-to-action button** that responds when users interact with it.

Task Description

- Add a button to a web page using HTML.
- Use AI to generate JavaScript that responds to the button click by:
 - Displaying a message or alert
- Ask AI to add meaningful comments explaining the JavaScript logic.

Expected Output

- A web page where:
 - Clicking the button triggers a visible response (alert or message)
 - JavaScript code is readable and commented
 - No page reload occurs



	tailored to your brand. This page says You have successfully signed up! OK the right audience. About Us We are a passionate team helping startups succeed in the digital world. Get In Touch Email us at hello@startup.example Sign Up Now 2026 StartUp. All rights reserved. tailored to your brand. modern technologies. the right audience. About Us We are a passionate team helping startups succeed in the digital world. Get In Touch Email us at hello@startup.example Signed Up Thank you for signing up! We will be in touch soon. 2026 StartUp. All rights reserved.	
	Task 3: Contact Form with Client-Side Validation Scenario	

You are developing a **contact form for a company website** to collect user queries safely and correctly.

Task Description

- Design a form with fields for:
 - Name
 - Email
 - Message
- Use AI to generate JavaScript validation logic that:
 - Prevents empty submissions
 - Checks for a valid email format
- Display clear error messages for invalid input.

Expected Output

- A functional form where:
 - Invalid input shows inline error messages
 - Valid input displays a success message
 - Validation happens on the client side using JavaScript

Get In Touch

Have a question? Send us a message and we'll get back to you.

Name

Email

Message

Hello There!!!

Send Message

	Have a question? Send us a message and we'll get back to you. Name Your name Email you@example.com Message Your message Send Message Thank you! Your message has been sent successfully.	
	Task 4: Dynamic List Management for a Product Page Scenario A product listing page needs to add or remove items dynamically without refreshing the page. Task Description • Create a simple HTML list to display items. • Use AI to generate JavaScript that: ◦ Adds a new item when an “Add” button is clicked ◦ Removes the last item when a “Remove” button is clicked • Ensure the DOM updates dynamically.	

	Expected Output • A web page where: ○ Items can be added and removed dynamically ○ Changes appear instantly without page reload ○ JavaScript logic is clear and maintainable	
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Product List

Add

Remove Last

- Website Builder
- SEO Toolkit
- Analytics Dashboard

Product List

Add

Remove Last

- Website Builder
- SEO Toolkit
- Analytics Dashboard
- laptop

	Product List Enter product name Add Remove Last Website Builder SEO Toolkit Analytics Dashboard	
	Task 5: Modal Popup for User Notifications Scenario You are building a user notification popup for announcements or alerts on a website. Task Description • Use AI to help generate: ◦ HTML structure for a modal popup ◦ CSS styling with a semi-transparent background overlay ◦ JavaScript to open and close the modal • Include multiple ways to close the modal (button or overlay click). Expected Output • A web page where: ◦ Clicking “Open Modal” displays the popup ◦ Popup overlays the page content clearly	

- Modal closes smoothly when instructed

Product List

AddRemove Last

Website Builder

SEO Toolkit

Analytics Dashboard

Open Modal

Product List

AddRemove Last

Website Builder

Notification

This is an important announcement! Check out our latest updates and new features now available on the platform.

Got it

Note: Report should be submitted a word document for all tasks in a

	single document with prompts, comments & code explanation, and output and if required, screenshots	
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